

Oluwatobi Melvyn Mayungbo

Data Scientist, Credit and Fraud Risk

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SUMMARY

Credit and fraud risk data scientist who builds the first working version. Built a credit decisioning system with calibrated default probabilities, cost-optimal cutoffs, reason codes and an audit trail, plus a fairness audit, and prototypes tools with AI coding assistants. Works in SQL and Python.

SKILLS

Payments and Risk: Credit Risk Decisioning, Transaction Fraud Detection, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

Statistics and Causal Inference: WOE Scorecards, Logistic Regression, Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting

Machine Learning: LightGBM, SHAP, Model Calibration (Isotonic Regression, Temperature Scaling), PyTorch, Scikit-learn, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Feature Engineering

Engineering and MLOps: SQL, Python, FastAPI, DuckDB, PostgreSQL, Redis, Docker, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, CI/CD (GitHub Actions), Git, Pytest, Streamlit, React

AI and LLM Systems: Prompt Engineering, LLM Evaluation and Guardrails, Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), Citation Verification, Model Context Protocol (MCP), Hugging Face, ONNX Runtime

Data Analysis: Pandas, NumPy, SQL, Anomaly Detection

PROJECTS

Credit Risk Decisioning and Fairness Audit System | [GitHub](#) | [Live App](#)

Tech: Python, LightGBM, scikit-learn, SHAP, WOE Scorecards, FastAPI, Docker

- Developed a full credit decisioning system producing calibrated default probabilities, cost-optimal approve and decline cutoffs, adverse action reason codes and an append-only audit trail that lets any individual decision be reconstructed months later.
- Benchmarked gradient boosting against a traditional WOE-binned logistic scorecard, prioritising calibration quality (Brier score, reliability diagrams, expected calibration error) over ranking metrics, because expected loss needs true probabilities rather than rankings.
- Audited fairness across sex, age, education and marital status using demographic parity and equal opportunity differences plus within-group calibration, and presented the accuracy and fairness tradeoff curve as a policy decision rather than silently optimising it away.

Sequence-Based Transaction Fraud Detection System | [GitHub](#)

Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP

- Benchmarked all architectures against a strong LightGBM baseline on engineered velocity and deviation features, reporting PR-AUC and precision at fixed review capacity (top 0.1%, 0.5%, 1%) across 3 seeds, with a per-attack-type breakdown covering card testing, account takeover and merchant compromise; sequence models cut expected cost per transaction 27% under an explicit cost model.

Reckon: Payment Reconciliation and Review System | [GitHub](#) | [Live App](#)

Tech: Python, PyTorch, DuckDB, FastAPI, React, Docker, Railway, PostgreSQL

- Designed a human-in-the-loop review interface in React showing match confidence scores, audit trails and suggested resolution actions to streamline exception handling.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.

- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.
- Internal Auditor, NISER**

May 2024 to July 2024
- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

- B.Sc. Statistics, University of Ibadan**

January 2021 to March 2025
- Professional Diploma in Artificial Intelligence, SQI College of ICT**

July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action